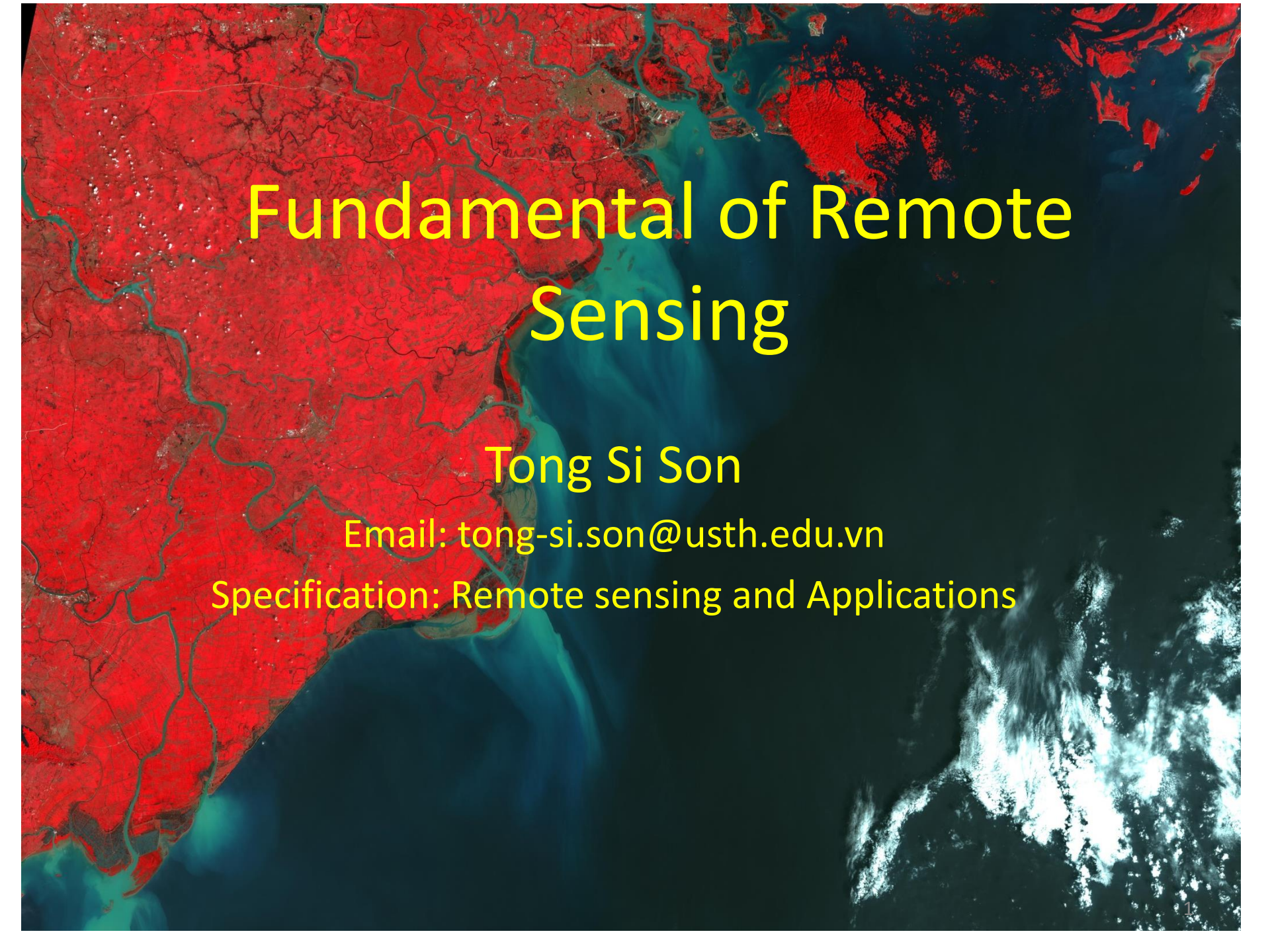


A satellite image of a coastal region. The land is colored in a vibrant red, showing intricate patterns of rivers, canals, and urban areas. The water is a deep blue, with some lighter blue areas indicating sediment or shallow depths. The sky is dark blue with some white clouds visible on the right side.

# Fundamental of Remote Sensing

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Specification: Remote sensing and Applications

# Practices

1. Exploring satellite data
2. Practice to measure the reflectance of surface object in the lab
3. Midterm presentation
4. Image pre-processing
5. Extract Information from satellite data

# 4. Image Pre-processing

1/ Calculate surface reflectance and Atmospheric correction

2/ Geometric correction

3/ Cutting image according to a border

4/ Image enhancement

# 4.1 Calculate surface reflectance and Atmospheric correction

- Reasons to use reflectance instead of radiance or DN
  - Comparing lab spectral curves and imagery spectral curves
  - Comparing imagery spectral from multi-temporal images
  - Comparing images from different sensors

$$\text{Spectral reflectance (\%)} = \rho_{\lambda} = \frac{\Phi_R}{\Phi_I}(\lambda)$$

# Calculate reflectance from measured radiance

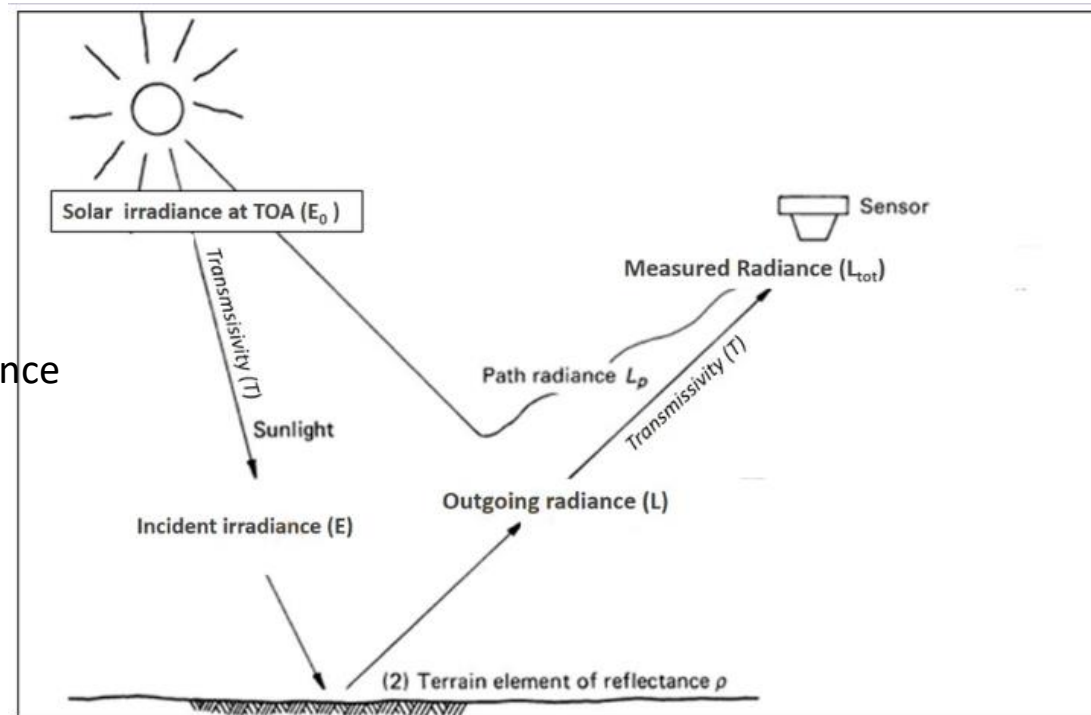
$$\text{Reflectance } \rho_{\lambda} = L/E$$

## 1. Estimate surface Irradiance (E)

- Correction of solar illumination properties
- Correction of incoming transmissivity

## 2. Estimate outgoing radiance (L)

- Estimation of  $L_{\text{tot}}$  by converting DN to radiance
- Correction of outgoing transmissivity (T)
- Subtract path radiance ( $L_p$ )
- Correct for Lambertian scattering



# Calculate reflectance from measured radiance

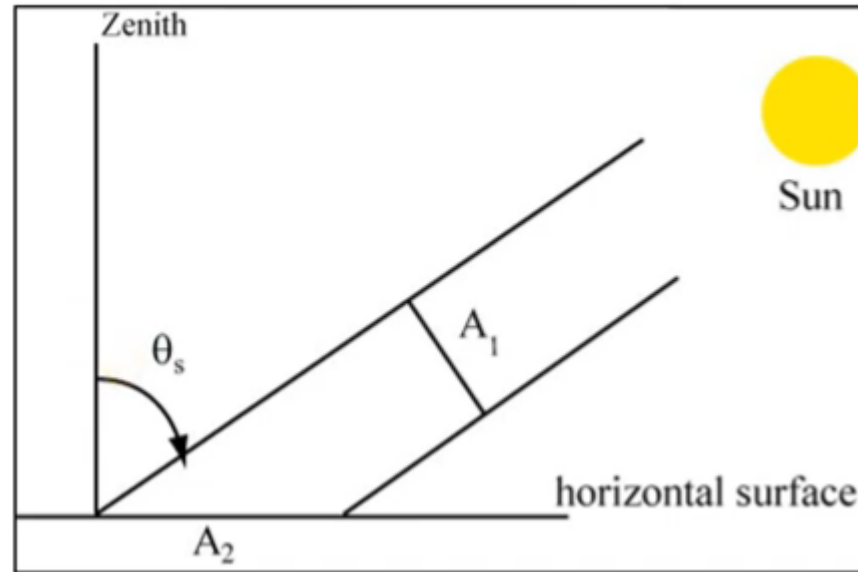
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$$E = E_0 * \cos\theta$$

$E_0$  = average irradiance at top of atmosphere (W/m<sup>2</sup>) ( $E_0$  solar constant = 1362 W/m<sup>2</sup>)

$E$  = instantaneous surface irradiance (W/m<sup>2</sup>)

$\theta$  = solar angle (90 – solar elevation angle)

# Calculate reflectance from measured radiance

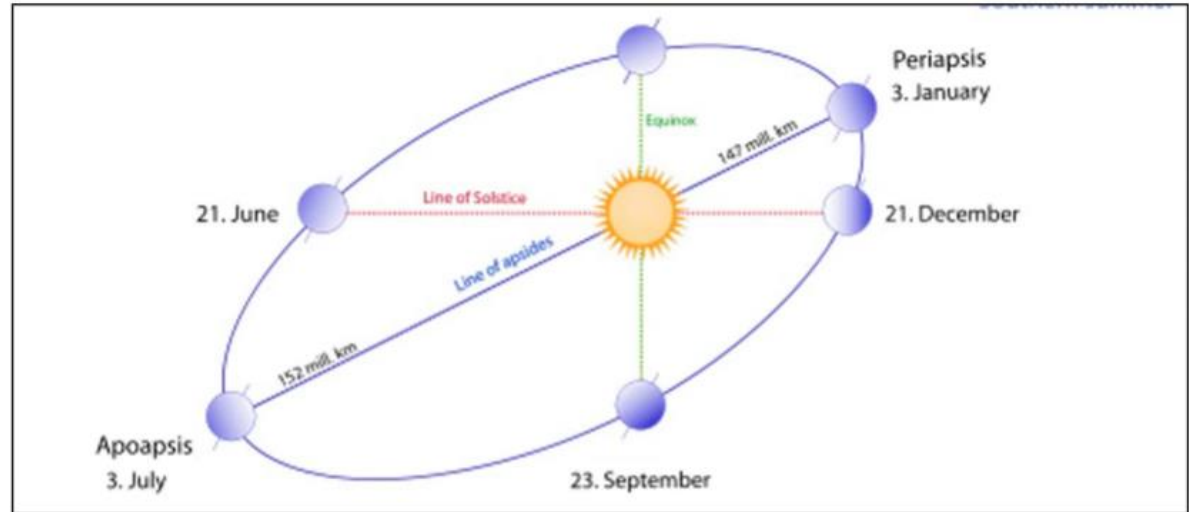
$$\text{Reflectance } \rho_{\lambda} = L/E$$

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## Correction of Earth-sun distance

*(sun energy decreases corresponding to  $1/d^2$ )*

$$E = (E_0 * \cos\theta) / d^2$$

Earth-sun distance:  $d = 0.983; 1.016$  astronomical unit (AU)

# Calculate reflectance from measured radiance

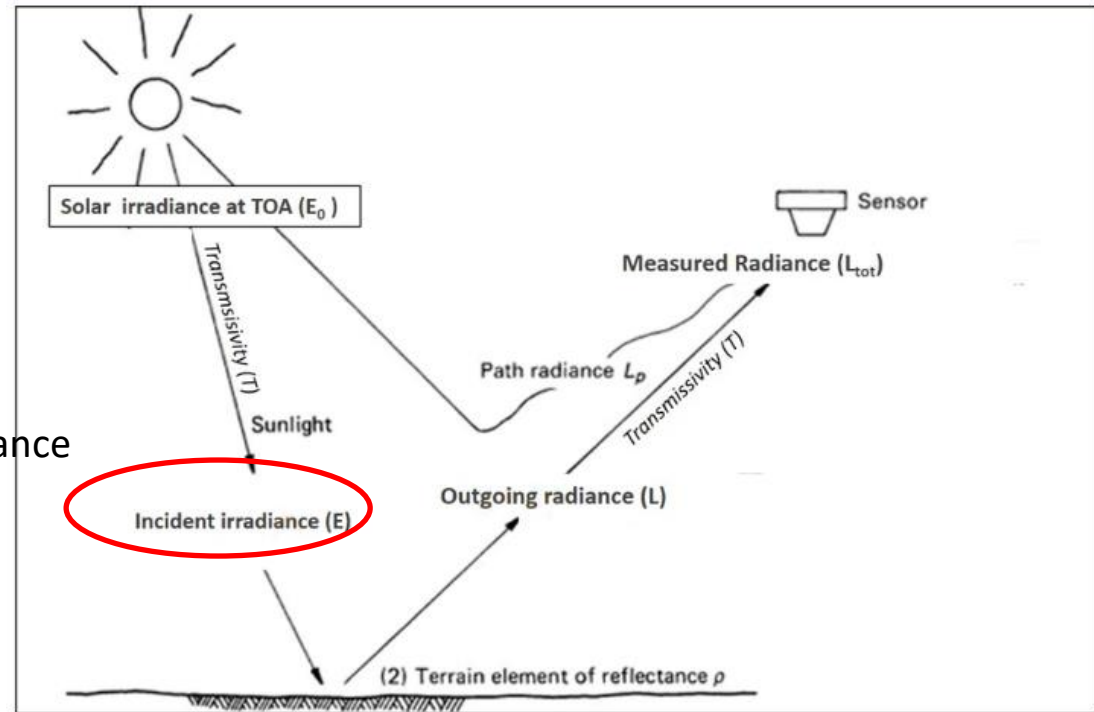
$$\text{Reflectance } \rho_\lambda = L/E$$

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## Correction of transmissivity (T model)

*(Transmissivity (T) = fraction of energy consumed on path through atmosphere (0-1))*

$$E = T * (E_0 * \cos\theta) / d^2$$

Earth-sun distance:  $d = 0.983; 1.016$  astronomical unit (AU)

# Calculate reflectance from measured radiance

$$\text{Reflectance } \rho_{\lambda} = L/E$$

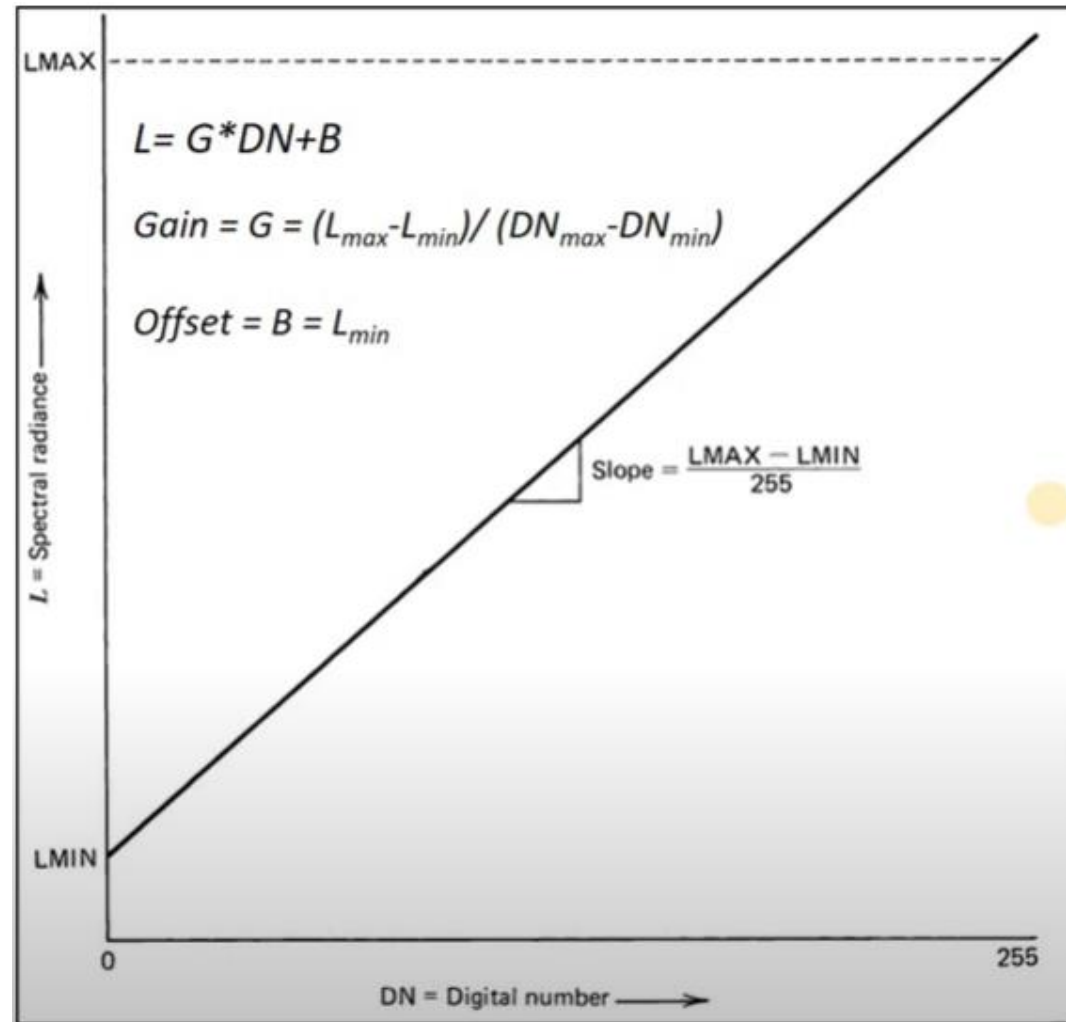
## 1. Estimate surface Irradiance (E)

- Correction of solar illumination properties
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**Estimate  $L_{\text{tot}}$  by converting DN's to radiance**



# Calculate reflectance from measured radiance

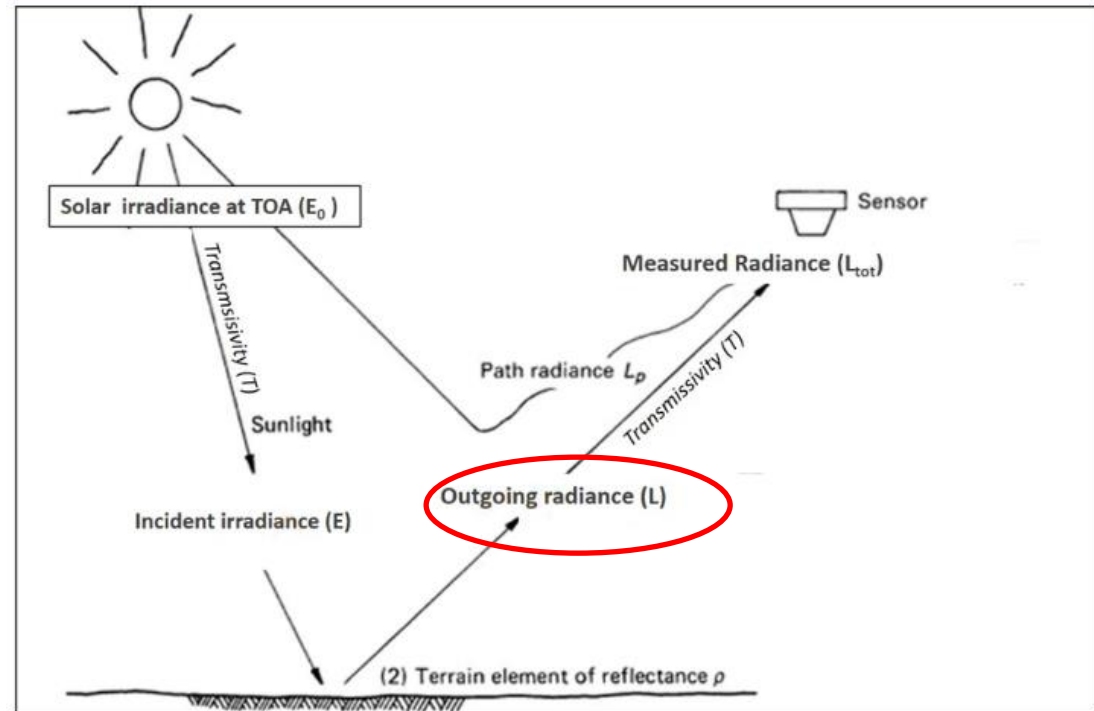
$$\text{Reflectance } \rho_{\lambda} = L/E$$

## 1. Estimate surface Irradiance (E)

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$$L_{\text{tot}} = L * T \Rightarrow L = L_{\text{tot}} / T$$

# Calculate reflectance from measured radiance

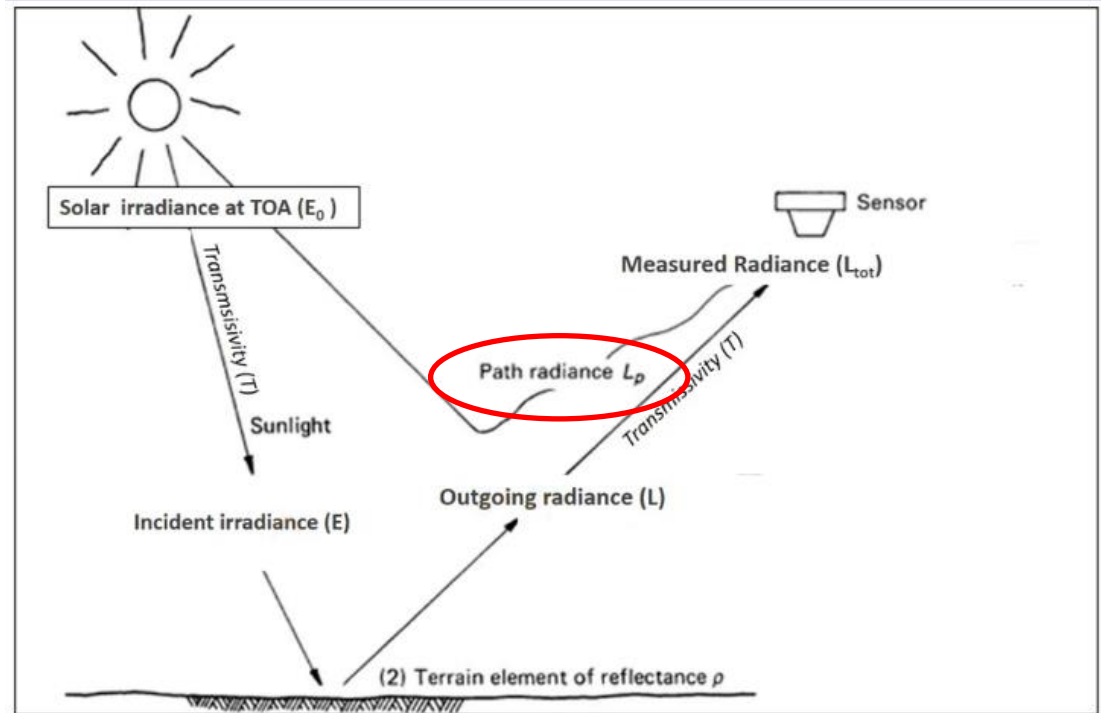
$$\text{Reflectance } \rho_{\lambda} = L/E$$

## 1. Estimate surface Irradiance (E)

- Correction of solar illumination properties
- Correction of incoming transmissivity

## 2. Estimate outgoing radiance (L)

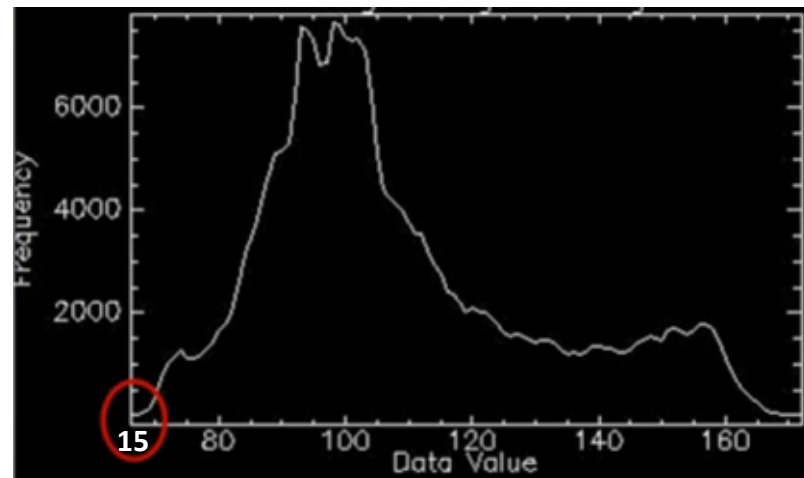
- Estimation of  $L_{\text{tot}}$  by converting DN to radiance
- Correction of outgoing transmissivity (T)
- Subtract path radiance ( $L_p$ )
- Correct for Lambertian scattering



$$L_{\text{tot}} = L * T + L_p$$

$L_p$  : caused by the emission, scattering from atmosphere;

*(Using method Dark Object Subtraction to detect the radiation of non reflectance surface)*



# Calculate reflectance from measured radiance

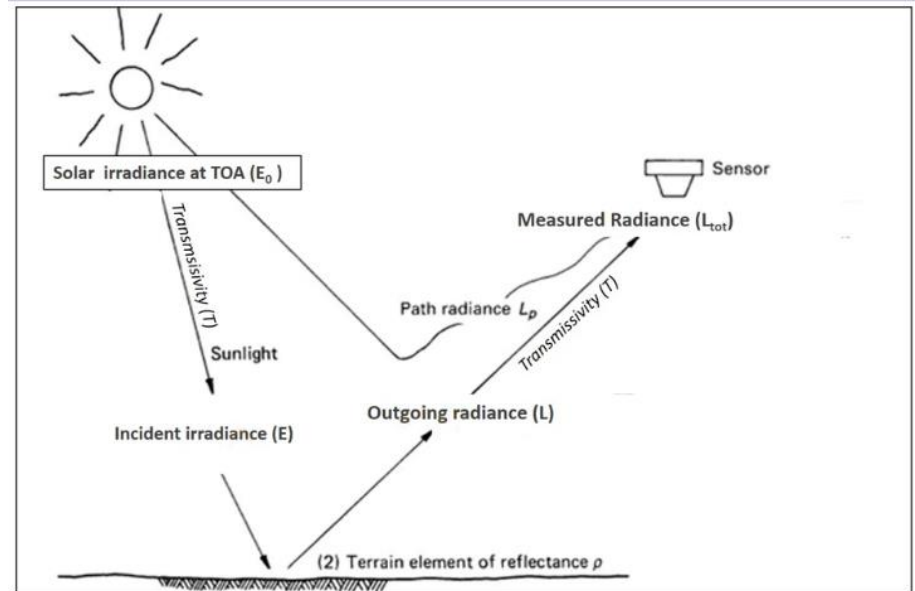
$$\text{Reflectance } \rho_\lambda = L/E$$

## 1. Estimate surface Irradiance (E)

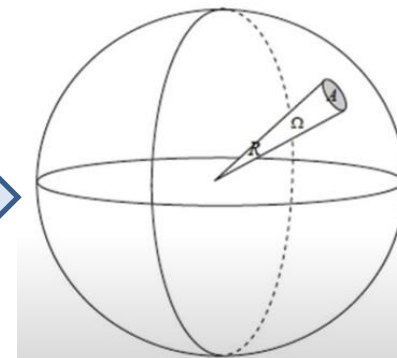
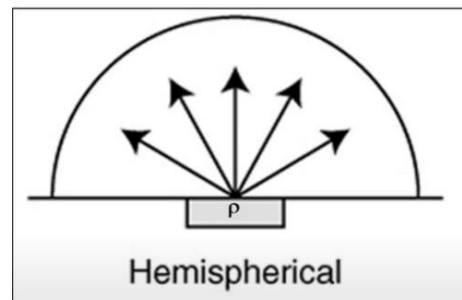
- Correction of solar illumination properties
- Correction of incoming transmissivity

## 2. Estimate outgoing radiance (L)

- Estimation of  $L_{\text{tot}}$  by converting DN to radiance
- Correction of outgoing transmissivity (T)
- Subtract path radiance ( $L_p$ )
- *Correct for Lambertian scattering (reflect according to a solid angle)*



$$L_{\text{tot}} = L * T / 2\pi + L_p$$



$$\text{Area of Sphere} = 4\pi R^2$$

$$\Omega = \frac{A}{R^2}$$

$$\text{Solid angle} = \Omega = 2\pi$$

# Calculate reflectance from measured radiance

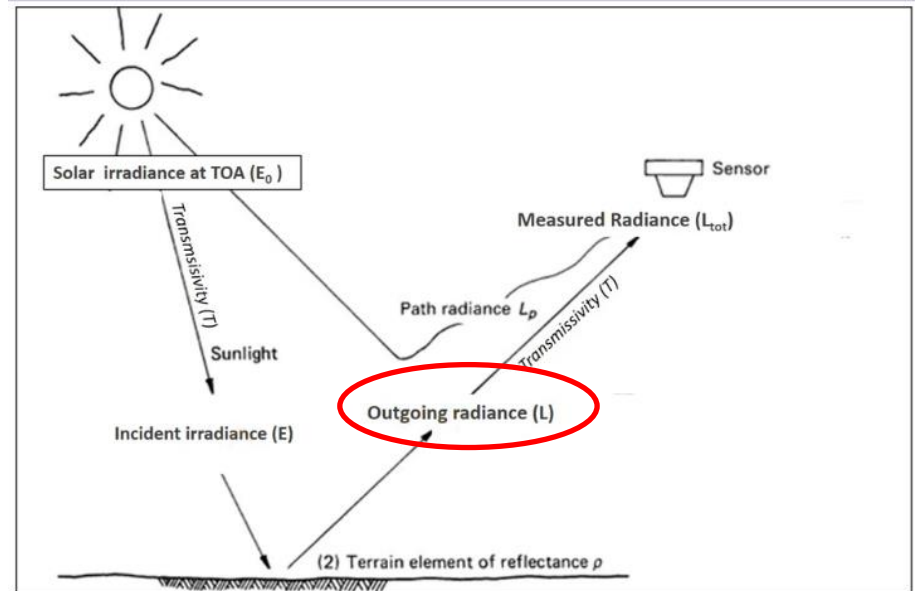
$$\text{Reflectance } \rho_{\lambda} = L/E$$

## 1. Estimate surface Irradiance (E)

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- Correction of incoming transmissivity

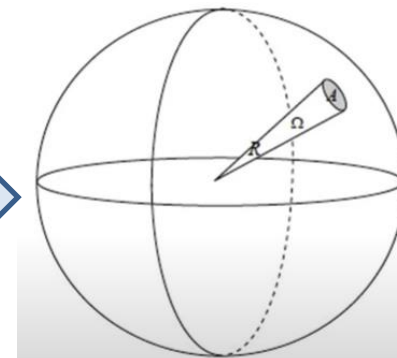
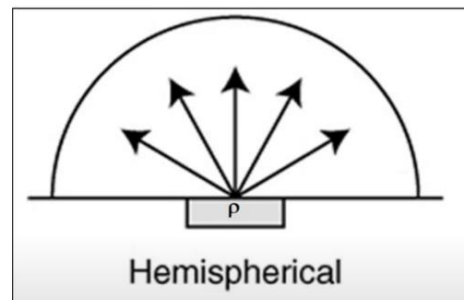
## 2. Estimate outgoing radiance (L)

- Estimation of  $L_{\text{tot}}$  by converting DN to radiance
- Correction of outgoing transmissivity (T)
- Subtract path radiance ( $L_p$ )
- *Correct for Lambertian scattering (reflect according to a solid angle)*



$$L_{\text{tot}} = L * T / 2\pi + L_p$$

$$L = (L_{\text{tot}} - L_p) * 2\pi / T$$



$$\text{Area of Sphere} = 4\pi R^2$$

$$\Omega = \frac{A}{R^2}$$

$$\text{Solid angle} = \Omega = 2\pi$$

# Calculate reflectance from measured radiance

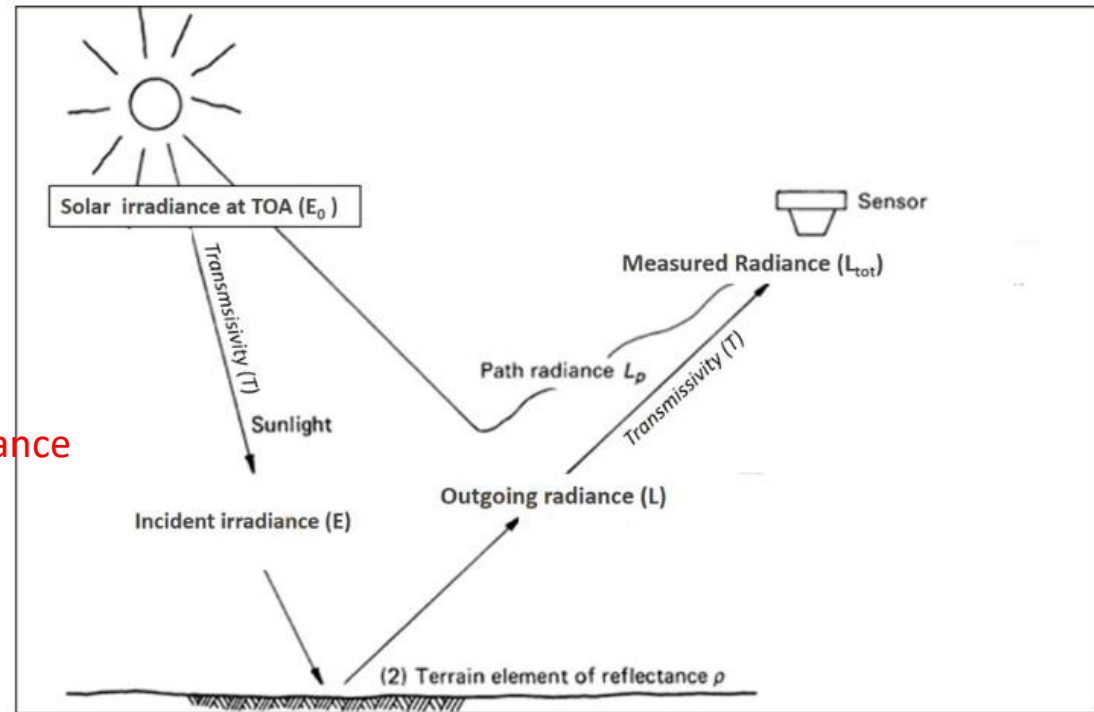
$$\text{Reflectance } \rho_{\lambda} = L/E$$

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- Correct for Lambertian scattering



$$\text{Reflectance } \rho_{\lambda} = L/E = [T * (E_0 * \cos\theta) / d^2] / [(L_{\text{tot}} - L_p) * 2 \pi / T]$$

# Calculate reflectance from measured radiance

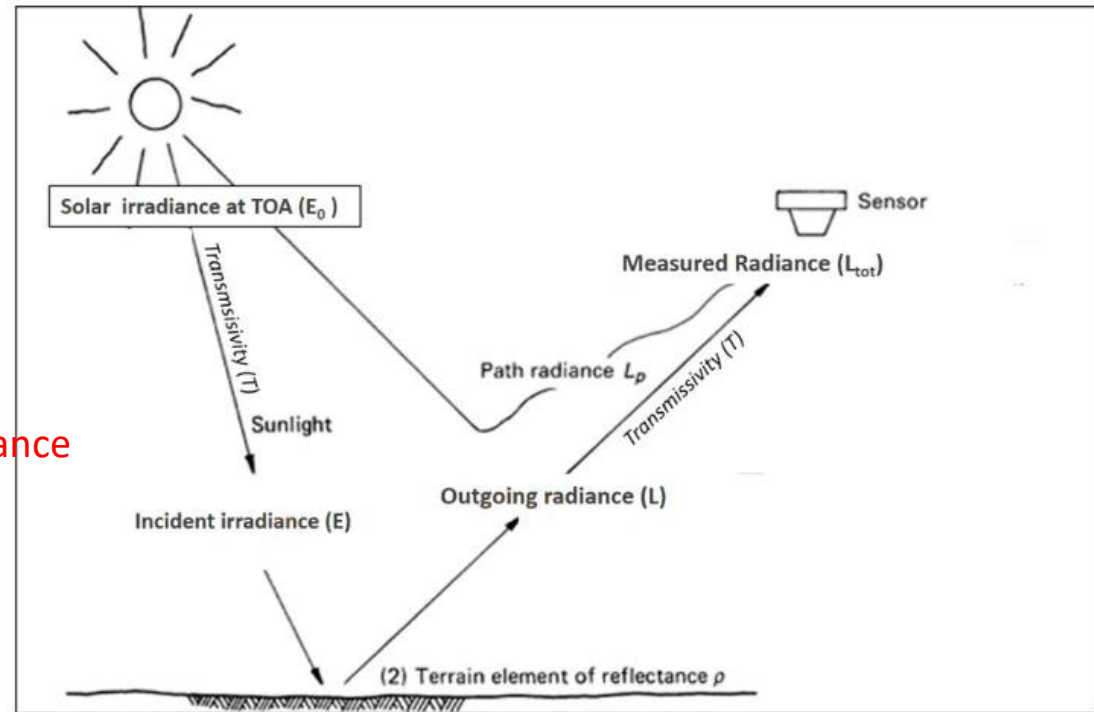
$$\text{Reflectance } \rho_{\lambda} = L/E$$

## 1. Estimate surface Irradiance (E)

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$$\text{Reflectance } \rho_{\lambda} = L/E = [T * (E_0 * \cos \Theta) / d^2] / [(L_{\text{tot}} - L_p) * 2 \pi / T]$$

# Equation Dark Object Subtraction (DOS) method

$$\text{REF} = \frac{(\text{PI} * \text{D} * \text{D} * (\text{Lsat} - \text{Lhaze}))}{(\text{Eo} * \text{Cos}(\text{TZ}) * \text{TAUz})}$$

- Lsat = The at-satellite radiance = the Top-Of-Atmosphere (TOA) radiance
- REF = Spectral reflectance (brightness) of the surface
- Lhaze = Atmospheric scattering for the given spectral band (i.e., path radiance) and is determined by the dark-object minimum DN values extracted from the image and is a CRITICAL parameter.
- Eo = Esun = Solar irradiance for the given spectral band (equal to the total amount of radiance coming in for the given band and is constant for the given band and sensor).
- D = Earth-sun distance in astronomical units (AUs) which is a function of time of year with a range from approximately 0.983-1.071
- TZ = Solar zenith angle which is equal to 90 minus sun elevation angle ---I referred to this as THETA.
- TAUz = Atmospheric transmittance along the path from the sun to the ground surface (atmospheric absorption). The COST model sets this equal to Cos(TZ) and was the first image based model to not ignore this parameter (i.e., set it equal to 1).
- PI = 3.1416

# Calculate sensor spectral radiance

$$L_{\lambda} = M_L * Q_{cal} + A_L$$

$L_{\lambda}$  = Spectral radiance (W/(m<sup>2</sup> \* sr \*  $\mu$ m))

$M_L$  = Radiance multiplicative scaling factor for the band (RADIANCE\_MULT\_BAND\_n from the metadata).

$A_L$  = Radiance additive scaling factor for the band (RADIANCE\_ADD\_BAND\_n from the metadata).

$Q_{cal}$  = L1 pixel value in DN

- Collect reflectance information of surface objects on images and compare with the theoretic reflectance

(Apply for TOA Landsat data)

## Convert DN to radiance

$$L_{\text{sat}} = \text{DN} * \text{RADIANCE\_MULT} + \text{RADIANCE\_ADD}$$

RADIANCE\_MULT: In metadata

RADIANCE\_ADD: In metadata

D = Earth-sun distance: In metadata

TZ = Solar zenith angle: : In metadata

| Esun: | Landsat 8 |      |      |        |            | ESUN  |
|-------|-----------|------|------|--------|------------|-------|
|       | Band      | Low  | High | Center | Linear     | TOA*  |
|       | 2 (blue)  | 0.45 | 0.51 | 0.480  | 1991 * *   | 2067  |
|       | 3 (green) | 0.53 | 0.59 | 0.560  | 1812 * * * | 1893  |
|       | 4 (red)   | 0.64 | 0.67 | 0.655  | 1549       | 1603  |
|       | 5 (NIR)   | 0.85 | 0.88 | 0.865  | 962.6      | 972.6 |
|       | 6 (SWIR)  | 1.57 | 1.65 | 1.610  | 251.7      | 245.0 |
|       | 7 (SWIR)  | 2.11 | 2.29 | 2.200  | 86.30      | 79.72 |
|       | 9 (SWIR)  | 1.36 | 1.38 | 1.370  | 385.4      | 399.7 |
|       | 1 (Blue)  |      |      |        |            | 1895  |

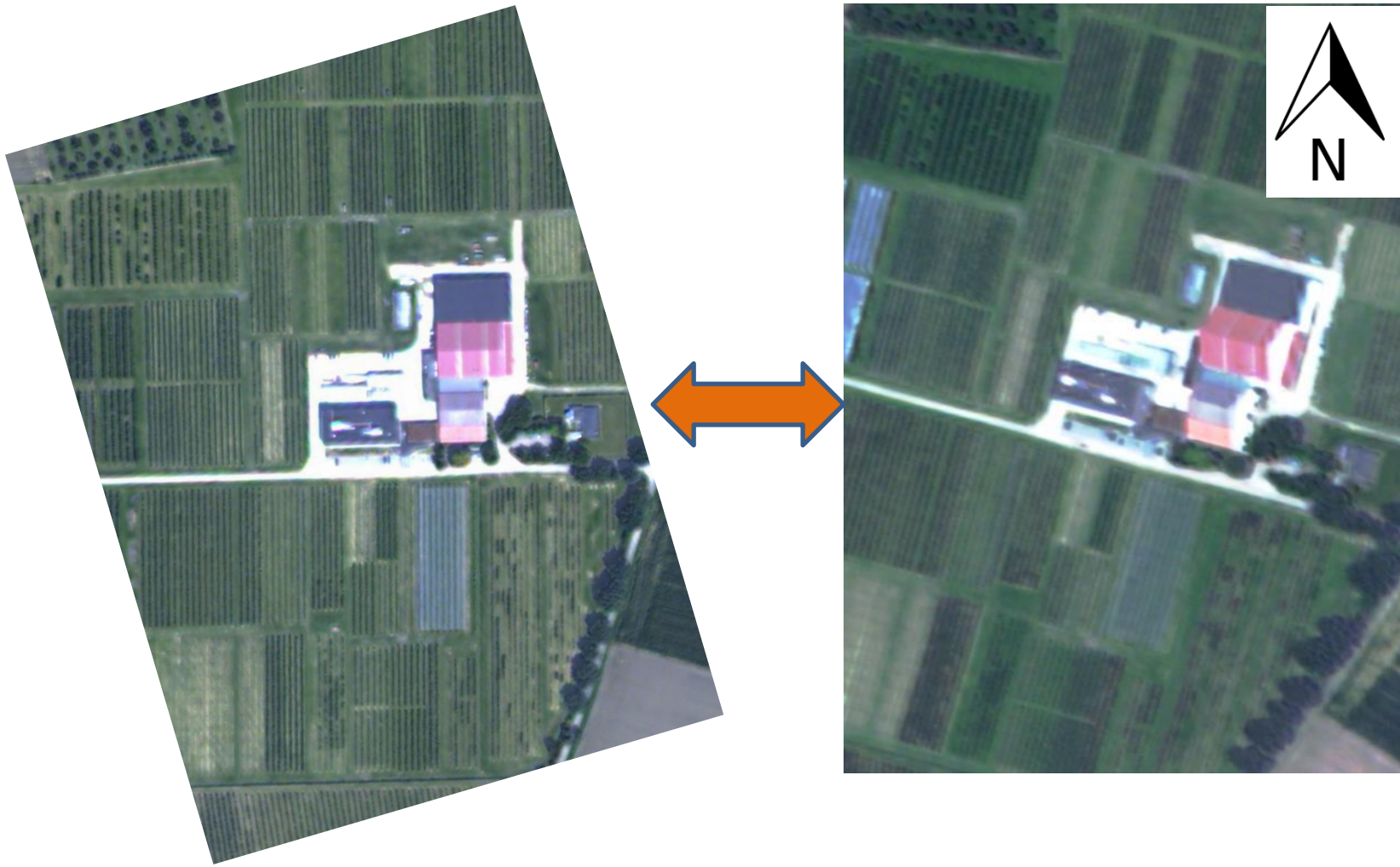
Lhaze = Atmospheric scattering ? B2; 42

# Steps to calculate surface reflectance

1. Open separated bands
  - Raster/ Geometric/ Colocation (apply Band 1 to Band 6)
  - Change the name and wavelength of Bands
  - Spectrum view to observe the variation of spectral variation
2. Convert DN to Radial
  - In SNAP: Raster/Band math*
3. Calculate Lhaze
  - Detect Radial at the dark body (Clear lake), or using histogram to determine the min of dark object*
4. Running the DOS equation
  - In SNAP: Raster/Band math*
5. Recalculate for all the bands
6. Comparing the image before and after correction
  - (should convert to SNAP format to see the Spectrum view)*

## 4.2 Geometric correction (Using QGIS)

- Install QGIS and Google Earth
- Removal of spatial sensor distortions and link it known geographic coordinate system
- Registration >>> matching two (or more) images to each other
- Field measurements, GIS data



# Image warping

Ground control points (GCP)

## Properties

- Stability
- Extractability
- Frequency
- Evenly distributed
- Output coordinates known



# STEPS

1/ Open image in QGIS

*QGIS: Raster/Georeferencer*

2/ Open Google Earth

3/ Determine GCP on Google Earth and QGIS

4/ Run Geometric correction

*Rater/ Georeferencer*

## 4.3 Cutting image according to a border

### Using SNAP

1/ Open Geometric, atmospheric corrected image.

2/ In SNAP: Import shape file

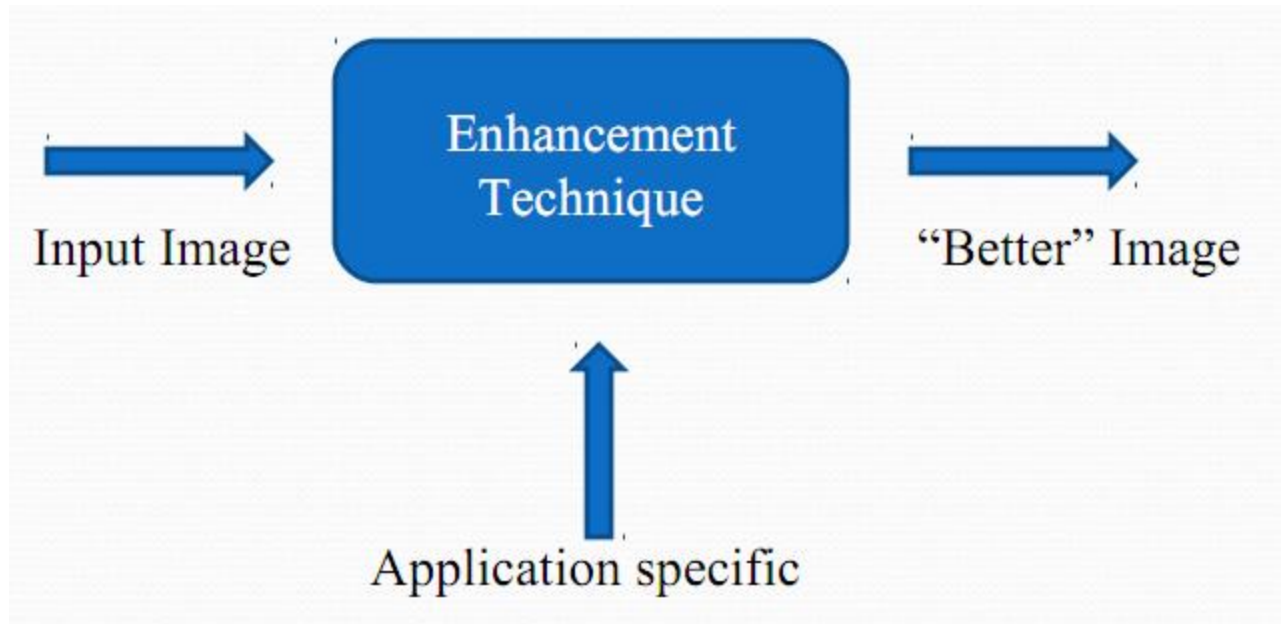
*Vector/Import*

3/ Clip raster following a vector

*In SNAP: Raster/ Masks/ LandSea mask*

# 4.4 Image enhancement

- **Image enhancement process** consists of a collection of techniques that seek to improve the visual appearance of an image or to convert the image to a form better suited for analysis by a human or machine.
- **The principal objective** of image enhancement is to modify attributes of an image to make it more suitable for a given task and a specific observer.



Thank you for your attention!